

Is Retail Sentiment Spanned by the Fama–French Factors?

A Complementary-Factor Analysis of Signed Abnormal Turnover in the STOXX Universe

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Abstract

In a companion paper we show that Signed Abnormal Turnover (SAT) — a European retail sentiment proxy grounded in [Baker and Stein \(2004\)](#) — does not *predict* next-month returns in the STOXX universe once look-ahead bias is removed. This paper asks the question that survives the null: does retail sentiment capture a distinct source of *contemporaneous* common return variation that the standard factor model misses? We construct the Retail Sentiment Factor (RSF) exactly as Fama and French construct SMB and HML: an independent 2×3 double sort on size and SAT, value-weighted, formed at month-end t and held over month $t+1$. Over 185 months (2010–2025) RSF earns 0.12% per month (annualised Sharpe 0.29). The central result is a spanning regression: RSF regressed on the European FF5 + momentum factors leaves an unspanned alpha of 0.22% per month ($t = 2.24$, $p = 0.025$) with $R^2 = 0.03$, and no pairwise correlation with an FF factor exceeding 0.12 in absolute value. The Fama–French factors do not span retail sentiment. Adding RSF as a seventh right-hand-side factor to regressions of the 25 PE/SAT test portfolios leaves their alphas essentially unchanged, indicating that the sentiment factor is a genuine *complement* to the FF regressors rather than a repackaging of test-asset mispricing. We conclude that retail sentiment belongs on the right-hand side of European factor models — something to be stripped from returns — not in a forecasting rule. Honest caveats: the spanning alpha clears the conventional $t = 2$ bar but not the [Harvey et al. \(2016\)](#) threshold of 3.0, and the post-surge sample is short.

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1 Introduction

A factor does not need to forecast returns to matter for asset pricing. None of the Fama–French factors is a return forecaster in the time series; their role is cross-sectional and contemporaneous. SMB is long small stocks and short big stocks *this month*; its factor premium is the average of that contemporaneous long–short return; and its use in applied work is as a right-hand-side regressor that strips size-related comovement out of whatever is being evaluated — a fund, a strategy, an anomaly. The economically meaningful question about a candidate factor is therefore not “does it predict?” but “is it spanned?": can its average return and its variation be reproduced by a portfolio of the existing factors (Barillas and Shanken, 2017; Fama and French, 2018)?

This paper asks that question for European retail sentiment. In a companion paper we construct Signed Abnormal Turnover (SAT) — abnormal trading volume signed by return direction, the retail sentiment proxy available where the US BJZZ identification (Boehmer et al., 2021) is not — and show that under strictly correct sort/holding alignment it does not predict next-month STOXX returns: quintile spreads, Fama–MacBeth slopes, and factor-model alphas are all statistically and economically null. That result closes the forecasting door but leaves the spanning door open. Retail sentiment moves prices contemporaneously and observably: coordinated retail flows generated some of the largest single-name price events of the post-2020 era. If those flows generate *common* variation across many stocks — a sentiment “wave” that lifts high-attention names together — then a portfolio sorted on sentiment intensity constitutes a risk factor in the ordinary sense, whether or not the wave is forecastable.

We test this directly. RSF is built with the exact Fama–French 2×3 methodology: at each month-end, stocks are independently split by size (median) and SAT (30th/70th percentiles); six value-weighted portfolios are held over the following month; RSF is the average high-SAT return minus the average low-SAT return, neutralised across size legs. The size neutralisation matters — retail activity concentrates in smaller names, and without it a sentiment factor is substantially a size bet.

Three results follow. First, RSF earns a modest unconditional premium: 0.12% per month, annualised Sharpe 0.29. Second — the paper’s central finding — RSF is *not spanned* by the European FF5 + momentum factors: the spanning regression leaves an alpha of 0.22% per month ($t = 2.24$) with an R^2 of only 0.03, and RSF’s largest absolute correlation with any FF factor is 0.12. Retail sentiment is nearly orthogonal to the entire standard factor block. Third, adding RSF as a seventh factor to time-series regressions of the 25 PE/SAT-sorted test portfolios leaves their alphas essentially unchanged — consistent with the companion paper’s finding that those portfolios contain no sentiment-driven mispricing to explain, and reassuring on the concern that RSF merely repackages the test assets.

2 Why Spanning, Not Prediction

The distinction between predictive signals and pricing factors is worth making precise, because the retail-sentiment literature frequently blurs it.

A *predictive signal* is information at time t that forecasts returns over $t+1$: its value is a trading strategy, and the relevant test is an out-of-sample sorted-portfolio spread or a Fama–MacBeth slope on lagged characteristics. A *pricing factor* is a contemporaneous return series F_t that co-moves with test assets: its value is in explaining the cross-section of average returns and risk, and the relevant tests are spanning regressions among factors and reductions in test-asset alphas (Barillas and Shanken, 2017). The Fama–French factors pass the second kind of test, not the first.

The companion paper shows SAT fails the first test in Europe. This is informative but not disqualifying: momentum aside, most accepted factors would fail it too. The question here is whether sentiment-sorted portfolios carry common variation and average returns that the existing factors cannot reproduce. If yes, applied work on European equities — performance attribution, anomaly evaluation, risk decomposition — should include RSF on the right-hand side, exactly as it includes SMB, whether or not anyone can time it.

Remark 1 (On the phrase “retail sentiment factor”). *Throughout, RSF denotes the traded long–short return series defined in Section 4, not the SAT characteristic itself. The characteristic sorts the portfolios; the factor is the resulting return spread.*

3 Data and the SAT Characteristic

The stock panel is identical to the companion paper: STOXX Europe 600 constituents, June 2009 – December 2025, financials (GICS 40) excluded, Refinitiv EUR total returns, volume, shares outstanding, and market capitalisation; 478 stocks and 69,492 stock-months after eligibility screens. European FF5 + momentum factor returns and the risk-free rate come from the Ken French Data Library (daily Europe files aggregated to monthly).

SAT is constructed as in the companion paper: daily abnormal turnover is the out-of-sample residual of a rolling 252-day AR(2) on log turnover; daily SAT is abnormal turnover signed by the day’s return; monthly SAT is the within-month mean (minimum 10 valid days), winsorised at 1%/99% and z -scored cross-sectionally within each month. All information in month- t SAT is available at the close of month t .

4 Factor Construction

Definition 1 (RSF factor). *At the end of each formation month t , all eligible stocks are independently sorted by (i) market capitalisation, split at the cross-sectional median*

into Small (*S*) and Big (*B*), and (ii) monthly SAT, split at the 30th and 70th percentiles into Low (*L*), Mid, and High (*H*). The four corner portfolios (*S/H*, *B/H*, *S/L*, *B/L*) are value-weighted with month-*t* weights and held over month *t*+1. The factor return is

$$\text{RSF}_{t+1} = \frac{1}{2}(r_{t+1}^{SH} + r_{t+1}^{BH}) - \frac{1}{2}(r_{t+1}^{SL} + r_{t+1}^{BL}). \quad (1)$$

Months in which any corner leg holds fewer than five stocks are dropped.

This is the SMB/HML construction of [Fama and French \(1993\)](#) applied to the SAT characteristic. Two design points deserve emphasis. First, the size legs neutralise the mechanical small-cap tilt of retail activity: averaging within size halves means RSF is long high-sentiment names *relative to same-size peers*. Second, the same sort-at-*t* / hold-at-*t*+1 discipline enforced in the companion paper applies here: forward returns are taken only when the stock’s next observation is exactly one month ahead, and automated tests verify that a panel in which SAT drives only formation-month returns produces an identically zero factor.

Table 1: RSF Factor: Summary Statistics

Statistic	Value
Mean (%/month)	0.120
Std. dev. (%/month)	1.43
Annualised Sharpe ratio	0.29
Skewness	-0.17
Excess kurtosis	-0.14
Months	185

Table 1 reports the unconditional properties. The premium is modest — 0.12% per month, annualised Sharpe 0.29 — in line with what one should expect from a sentiment tilt after size neutralisation, and far from the implausible magnitudes that look-ahead contamination produces (see the companion paper’s diagnostic section).

5 Spanning Results

5.1 Correlations

Table 2: Pairwise Correlations of RSF with the FF5 + Momentum Factors

FF factor	Corr. with RSF
Mkt-RF	-0.066
SMB	+0.009
HML	+0.053
RMW	-0.120
CMA	+0.067
UMD	-0.049

Table 2 is the first-pass evidence: no correlation exceeds 0.12 in absolute value. RSF is close to orthogonal to the market, both sides of the size and value sorts, profitability, investment, and momentum. Whatever variation the sentiment sort captures, it is not a linear repackaging of an existing factor.

5.2 The Spanning Regression

The formal test regresses the candidate factor on the incumbent factors ([Barillas and Shanken, 2017](#); [Fama and French, 2018](#)):

$$\text{RSF}_t = \alpha + \beta_1 \text{MktRF}_t + \beta_2 \text{SMB}_t + \beta_3 \text{HML}_t + \beta_4 \text{RMW}_t + \beta_5 \text{CMA}_t + \beta_6 \text{UMD}_t + \varepsilon_t. \quad (2)$$

Under the null that the FF factors span RSF, $\alpha = 0$: a mean–variance investor already holding the six factors gains nothing from adding RSF. A significant α means RSF expands the mean–variance frontier of the factor set.

Table 3: Spanning Regression: RSF on the European FF5 + Momentum Factors

Variable	Coefficient	t -stat	p -value
α	0.220%**	2.24	0.0249
Mkt-RF	-0.034	-1.26	0.2091
SMB	0.007	0.12	0.9073
HML	-0.070	-0.99	0.3245
RMW	-0.179**	-2.04	0.0410
CMA	0.032	0.30	0.7655
UMD	-0.040	-0.91	0.3650
R^2	0.031		
N (months)	185		

Notes: Monthly observations; Newey–West standard errors with 6 lags. The alpha is expressed in percent per month. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The unspanned alpha is 0.22% per month ($t = 2.24$, $p = 0.025$) — almost double the raw premium, because RSF’s small negative loadings on the market and profitability factors actually detract from its raw mean. The R^2 of 0.03 says the six factors jointly explain three percent of RSF’s variance: spanning fails not marginally but comprehensively. In the [Barillas and Shanken \(2017\)](#) framework, this is the entire case for adding a factor — the candidate’s alpha against the incumbents is the only statistic that matters for model comparison.

5.3 Does RSF Explain the Test Assets?

The complementary question is what RSF does on the right-hand side. We re-estimate the time-series regressions of the 25 PE/SAT test portfolios from the companion paper, first on the FF6 factors, then on FF6 + RSF.

Table 4: Test-Asset Alphas: FF6 versus FF6 + RSF (25 PE/SAT Portfolios)

Model	Mean $ \alpha $ (%/month)	$\# t > 2$	Mean R^2
FF6	0.336	2	0.510
FF6+RSF	0.351	3	0.523

Table 5: RSF Loadings of the 25 PE/SAT Portfolios (FF6 + RSF Model)

PE Quintile	SAT Quintile				
	Q1 (Bear)	Q2	Q3	Q4	Q5 (Bull)
Q1 (Low PE)	-0.52	-0.13	-0.25	0.18	0.56
Q2	-0.36	-0.08	-0.02	0.31	0.43
Q3	-0.44	-0.17	0.16	-0.12	0.26
Q4	-0.46	-0.16	-0.08	0.47	0.33
Q5 (High PE)	-0.74	-0.39	-0.15	0.33	0.56

Notes: Slope coefficients on RSF from time-series regressions of each portfolio’s excess return on the six FF factors plus RSF.

Two observations. First, mean absolute alphas are essentially unchanged when RSF is added (Table 4) — as they must be, given the companion paper’s finding that these portfolios contain no sentiment-related mispricing after correct alignment. RSF is not manufacturing its spanning alpha by absorbing test-asset alphas. Second, the RSF loadings in Table 5 line up with the SAT sort: high-SAT columns load positively, low-SAT columns negatively. The factor covaries with the characteristic that defines it, in the cross-sectional pattern one requires of a genuine common factor (Fama and French, 2015).

6 Interpretation and Caveats

6.1 What RSF Is

The results support a specific and limited claim: sentiment-sorted European portfolios carry a source of common return variation that is close to orthogonal to the FF5 + momentum block and earns a premium the incumbents cannot reproduce. The natural applications are on the right-hand side:

1. **Performance attribution.** A European fund or strategy tilted toward high-attention retail names will show FF6 alpha that is in fact sentiment-factor exposure. Adding RSF separates the two.
2. **Anomaly evaluation.** New European anomalies should demonstrate alphas robust to RSF, precisely because sentiment comovement is unspanned by the standard block.
3. **Risk decomposition.** RSF provides a measurable sentiment-wave exposure for risk models covering the post-2020 European retail era.

6.2 What RSF Is Not

RSF is not a trading signal: the companion paper’s null on predictability is unambiguous, and nothing here reverses it. Nor do we claim the premium is compensation for risk in a structural sense; with 185 months we cannot distinguish a risk premium from slow-moving mispricing of sentiment-exposed stocks.

6.3 Caveats

Three caveats bound the conclusion. First, the spanning alpha’s $t = 2.24$ clears the conventional bar but not the [Harvey et al. \(2016\)](#) threshold of 3.0 appropriate for new factor claims; on that stricter standard, RSF is a promising candidate rather than an established factor. Second, the sample is one universe (STOXX) and one construction (2×3 size/SAT); replication in other European universes and with alternative breakpoints is the obvious next step. Third, the post-surge subsample — where the mechanism should be strongest — is short; a regime-split spanning test will not be well-powered until the post-2020 era is longer.

7 Conclusion

Retail sentiment in European equities does not forecast returns, but it is not redundant either. Constructed with the standard Fama–French 2×3 methodology and strict formation/holding alignment, the Retail Sentiment Factor earns 0.12% per month, is almost uncorrelated with every FF5 + momentum factor, and leaves a spanning alpha of 0.22% per month ($t = 2.24$) against the full six-factor block with an R^2 of 0.03. The Fama–French factors do not span retail sentiment. The constructive conclusion for applied work is to treat RSF the way SMB and HML are treated: as a contemporaneous regressor that strips a distinct dimension of common variation out of returns — not as something to predict with, but as something to control for.

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